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High Performance Processors and Systems

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Name
Last Name

Problem 1 (4 points)	
Problem 2 (6 points)	
Problem 3 (6 points)	
Question 1 (6 points)	
Question 2 (6 points)	
Question 3 (4 points)	
Total (32 points)	

To pass the exam (the following conditions are in AND):

- The overall score has to be equal or higher than 18
- At least 1 problem has to be correct (at least it has to be evaluated with 85% of the “available” points)
- At least 9 points from the problems
- At least 9 points from the questions

Problem 1

Consider two CPUs: CPU1 e CPU2. CPU1 has clock cycle of 2 ns while CPU2 has an operating frequency of 700MHz. Given the following frequencies of occurrence of the instructions for the two CPUs

Operation type	Frequency	CPU1	CPU2
A	0.3	3	1
B	0.1	1	4
C	0.2	2	1
D	0.3	1	3
E	0.1	2	1

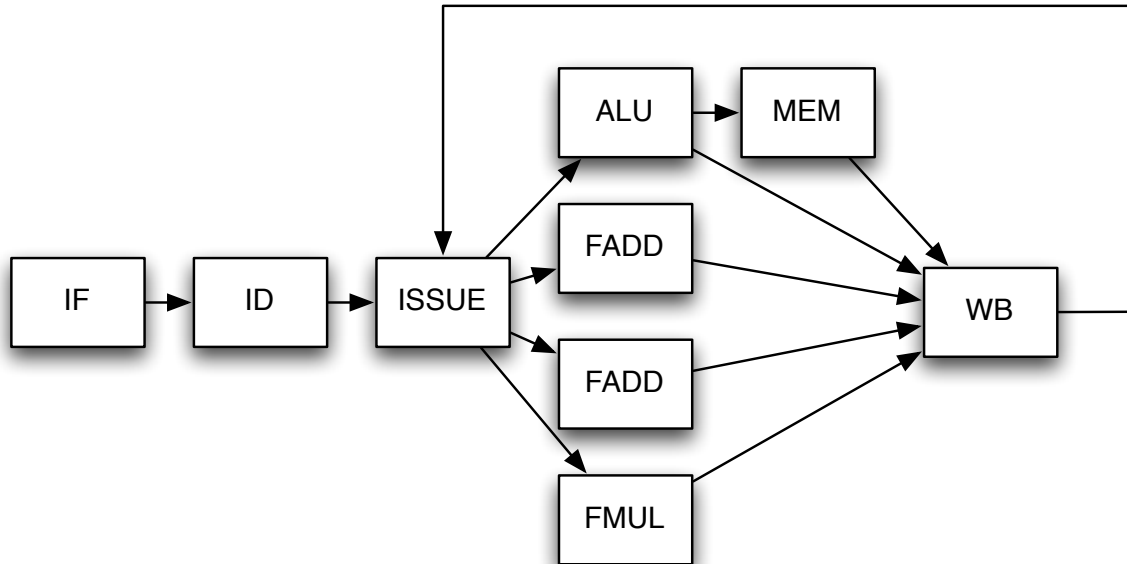
- a) Compute the average CPI for CPU1 and CPU2
- b) Which is the slowest CPU?
- c) Assume that the fastest CPU is modified in such a way that:
 - All instructions require 7 clock cycles;
 - Memory stalls are not considered;
 - The miss penalty is 11 clock cycles

Assuming a miss rate of 13% and assuming 5 memory references on average for each instruction determine the impact on performance of the cache with respect to an ideal cache?

Answer

Problem 2

In this problem we will examine the execution of a code segment on the following single-issue out-of-order processor:



You can assume that:

- All functional units are pipelined
- ALU operations take 1 cycle
- Memory operations take 2 cycles (includes time in ALU)
- Floating-point add instructions take 4 cycles
- Floating-point multiply instructions take 5 cycles
- There is no register renaming
- Instructions are fetched, decoded and issued in order
- The issue stage is a buffer of unlimited length that holds instructions waiting to start execution
- An instruction will only enter the issue stage if it does not cause a WAR or WAW hazard
- Only one instruction can be issued at a time, and in the case multiple instructions are ready, the oldest one will go first

To simplify book-keeping for this problem we will only track instructions during a few critical stages:

- **I** - When an instruction enters the issue stage
- **E** - When an instruction starts execution (enters FU)
- **C** - When an instruction completes execution (enters WB, results available this cycle)

For this problem we will be using the following code:

I ₁	L.D	F0, 0(R0)
I ₂	MUL.D	F0, F1, F1
I ₃	ADD.D	F2, F1, F1
I ₄	ADD.D	F4, F3, F3
I ₅	ADDI	R1, R3, 8
I ₆	ADD.D	F6, F0, F0

Problem 1.A

Fill in the following table to indicate how the given code will execute. Assume all register values are available at the start of execution.

[illegible]

Problem 1.B

The following new instruction has to be added to the previously reported piece of code: **I7 MUL.D F9, F6, F9**

```
I1    L.D      F0, 0(R0)
I2    MUL.D   F0, F1, F1
I3    ADD.D   F2, F1, F1
I4    ADD.D   F4, F3, F3
I5    ADDI    R1, R3, 8
I6    ADD.D   F6, F0, F0
I7    MUL.D   F9, F6, F9
```

- 1) Can you reschedule the code to achieve better performance?
- 2) Evaluate how many clock cycles are necessary to execute the whole piece of code (consisting of all the 7 instructions) and explain the reason.

Problem 3

Assume that the following code is executed on a CPU with TOMASULO and with the following units:

- 1 LOAD/STORE unit with Latency equal to 1
- 2 MULT units with latency equal to 10
- 3 ADD/SUBD unit with latency equal to 2

And the following information:

First load 8 clock cycles (cache miss)

Following loads 1 clock cycle (hit)

$$R1 = 20$$

Show the computation steps of the following lines of code:

```
Loop:    LD F0 0 R1
         ADDD F6 F4 F0
         MULTD F10 F2 F4
         ADDD F4 F10 F2
         SUBI R1 R1 #20
         BNEZ R1 Loop
```

[illegible]

Tables

[illegible]

t =				R1 =						
Name	Op	Vj	Vk	Qj	Qk	Etime		Busy	Addr	
mult1							Load			
mult2										
add1										
add2										
add3										
F0	F2	F4	F6	F10	F12					

[illegible]

[illegible][illegible][illegible]

[illegible][illegible][illegible]

t =				R1 =				
Name	Op	Vj	Vk	Qj	Qk	Etime		Busy
mult1							Load	
mult2								
add1								
add2								
add3								
F0	F2	F4	F6	F10	F12			

[illegible][illegible]

t =				R1 =				
Name	Op	Vj	Vk	Qj	Qk	Etime		Busy
mult1							Load	
mult2								
add1								
add2								
add3								
F0	F2	F4	F6	F10	F12			

[illegible][illegible]

t =				R1 =				
Name	Op	Vj	Vk	Qj	Qk	Etime		Busy
mult1							Load	
mult2								
add1								
add2								
add3								
F0	F2	F4	F6	F10	F12			

t =				R1 =					
Name	Op	Vj	Vk	Qj	Qk	Etime		Busy	Addr
mult1							Load		
mult2									
add1									
add2									
add3									
F0	F2	F4	F6	F10	F12				

Name	Op	Vj	Vk	Qj	Qk	Etime
mult1						
mult2						
add1						
add2						
add3						

F0	F2	F4	F6	F10	F12

t =			R1 =					
Name	Op	Vj	Vk	Qj	Qk	Etime		
mult1							Load	Busy
mult2								Addr
add1								
add2								
add3								
F0	F2	F4	F6	F10	F12			

Name	Op	Vj	Vk	Qj	Qk	Etime
mult1						
mult2						
add1						
add2						
add3						

t =			R1 =						
Name	Op	Vj	Vk	Qj	Qk	Etime		Busy	Addr
mult1							Load		
mult2									
add1									
add2									
add3									
F0	F2	F4	F6	F10	F12				

[illegible]

t =				R1 =				
Name	Op	Vj	Vk	Qj	Qk	Etime		
mult1							Load	
mult2								
add1								
add2								
add3								
F0	F2	F4	F6	F10	F12			

[illegible]

Question 1

In real processors, some architectural limits do not allow full exploitation of the ILP present in the program. Specify these architectural limits and specify how they impact on the potential ILP.

Question 2

IN MIMD machines specify the type of memory organization available and how to choose among them. Specify then the possible memory address space models available, their advantages and disadvantages and their relation to the memory organization, if any.

Question 3

Why is a second or third level cache are useful? What kind of benefits it provides to the system performance? Explain how it impacts by showing the Average memory access time formula.

What other technique of cache improvement can provide a solution to the same problem?

